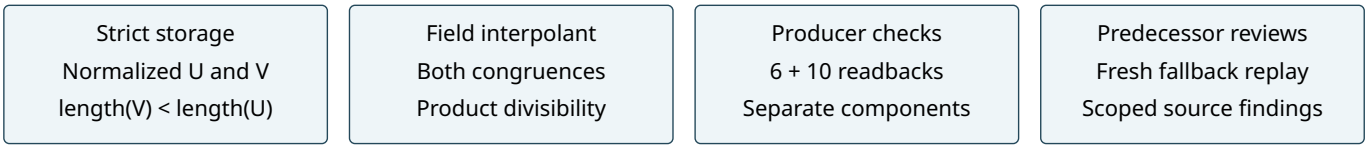
Strict storage and field CRT interpolation

Two new producer components plus two predecessor reviews, 2026-11-01. New component reviews pending.



Strict storage result. Four strengthened declarations expose normalized output lists and the strict inequality length(V)<length(U) through the reduction step, initialized loop, reduction entry and full generic fallback endpoint. They preserve the predecessor assumptions, actual quotient/remainder calls and original Run relation. The endpoint now exports every list-shape precondition of Divisor::from directly: nonempty U, at most three U coefficients, and strictly fewer V coefficients. It also retains canonicality, the two-coefficient V bound and decoded Mumford validity.

Why the interface changed. Independent review observed that array capacity and decoded validity alone do not imply the strict length assertion. U=[1], V=[0] fits the arrays and decodes to zero V but fails length(V)<length(U). The original construction had the strict remainder bound internally. The new declarations expose that fact without changing the input assumptions. Two controls record the capacity counterexample and correct identity storage U=[1], V=[]. Six modules and six named readbacks passed; the false theorem returned one; all bound inputs stayed unchanged. This is new producer work by the reviewer, not that author's independent review of these new declarations.

Field CRT result. A nonzero source norm supplies explicit polynomial Bezout coefficients proving coprimality of the two quadratic supports. The source interpolant V=Va+C(norm inverse)*(C(s0)+C(s1)*X)*Ua satisfies both ordinate congruences. Transporting the two input residual divisibilities through those congruences gives Ua*Ub dividing f-V squared. The field theorem needs norm nonzero, while the separate slope-nonzero guard is reserved for the optimized quotient-leading-coefficient argument.

Field CRT checks. Five main declarations, five fixed controls and ten named axiom readbacks passed, and the false theorem returned one. The full control uses Ua=X squared, Ub=X squared+1, Va=1, Vb=X+1 and f=X^5-X^4-X^3+1, with both input residual witnesses and V=1-X^3 checked. This is a polynomial-algebra fixture, with no curve smoothness claim. Zero-slope interpolation and zero-norm exclusion are separate controls. Retained failures concern an unavailable tactic, a missing Ruby executable and an ambiguous support name; final statements were not weakened.

Independent fallback review. The archived review binds commit 55b56b1e04 and reports faithful scoped source wiring. It independently recompiled six modules and checked nine named readbacks, the false theorem and three native checker rejection controls. Helper and standard-library/Mathlib caches were reused and hash-checked; this is not a fresh build of all dependencies. A first setup attempt omitted symlinked standard-library entries and failed before theorem checking; it is retained. The reviewer identified the strict-storage interface detail addressed by the new component above. The review remains evidence for the predecessor only.

Independent union/CRT review. The archived source review binds commit 929635eec1. It found no sign, orientation, subtraction-order or mutable-write mismatch within the eleven word-union and three CRT declarations. It inspected fifteen word-union and eight CRT readbacks but did not perform fresh reviewer replay. Its source-fidelity result does not establish forest termination, returned constraints, size-vector safety or divisor-class addition.

Assumptions and scope. Strict word storage retains prime p with 1<p<2^31, canonical coefficients, normalized nonzero input supports of degree at most two, degree-five f and both input residual divisibilities. Native correspondence is restricted to Field.n=1 and absent diagnostic tables. Field CRT algebra is over an arbitrary field and does not require smoothness or degree-five f. Native batching, compiler/copy semantics, optimized output equality and divisor-class interpretation remain separate work.

Sources and next steps. These components extend quintic-reduction-execution and union-and-crt-certificates at their pinned revisions. The stored plans and reports record the source and primary-literature context; standard polynomial Bezout/divisibility identities are reused. Next establish optimized quotient degree and leading coefficient under the slope guard, then bind the batched word arithmetic. The frozen 1,260-target scope and campaign deadlines are unchanged. No scientific run, timing or new cost ratio is included.